

SUMIT KUMAR KHADANGA

Cybersecurity Engineer · Digital Forensics & Malware Analysis

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PROFESSIONAL SUMMARY

Cybersecurity engineering student (B.Tech, graduating June 2026) with hands-on experience in memory forensics, malware analysis, and MITRE ATT&CK mapping. Built **WinVolAuto**, a memory-forensics automation tool (Volatility 3 + AI risk scoring) that **cut manual triage time and auto-mapped findings to ATT&CK techniques**, and **CTF Decoder**, a Bayesian adaptive decoding engine that **improved candidate-ranking accuracy across 30+ codecs**. Completed a cybersecurity internship covering vulnerability scanning, penetration testing, and red/blue team simulations. Seeking a **Digital Forensics / Malware Analyst / SOC Analyst Intern** role to apply offensive and defensive security skills in production environments.

Core Competencies: Digital Forensics · Memory Forensics · Malware Analysis · MITRE ATT&CK · Vulnerability Assessment · Penetration Testing · Network Security · OWASP Top 10 · Python

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C, SQL, HTML/CSS
Security: Memory Forensics (Volatility 3), Malware Analysis, MITRE ATT&CK, Rootkit Detection, OWASP Top 10
Tools: Wireshark, Nmap, Autopsy, Sleuth Kit, Burp Suite, Docker, Git, Kali Linux
Frameworks: FastAPI, Flask, React, Vue 3, PyQt6, Scikit-learn, SHAP, Streamlit
OS: Linux (Ubuntu, Kali), Windows

PROJECTS

WinVolAuto — AI-Augmented Memory Forensics Suite *Python · PyQt6 · Volatility 3*

- ▶ Automated end-to-end memory-dump triage across Windows, Linux, and macOS by building a GUI layer over Volatility 3 with dynamic plugin discovery and asynchronous execution (QThread), eliminating UI freezing on large (4GB+) dumps.
- ▶ Reduced manual ATT&CK-mapping effort by engineering an AI risk-scoring engine that auto-flags suspicious processes (code injection, hidden PIDs, network anomalies) and maps each finding directly to MITRE ATT&CK techniques.
- ▶ Accelerated investigation handoff by auto-generating structured PDF/JSON forensic reports with real-time plugin search across persistence, syscall-hook, and RAM-extraction triage modules.

CTF Decoder — Adaptive Decoding Engine *Python · FastAPI · Vue 3 · Bayesian ML*

- ▶ Built a self-learning decoding engine supporting 30+ codecs (Base64/32/85, XOR, ROT-N, Gzip, Bzip2, Morse, multi-layer payloads) using a Beta-Binomial Bayesian model combined with Shannon entropy and English-frequency scoring to rank decode candidates.
- ▶ Cut manual script review time by building a Code Script Solver that statically extracts and sandboxes dynamic execution of Python, C, and Java CTF scripts, auto-piping output through the decoder for flag detection.
- ▶ Shipped to production with zero added infra cost by deploying as a Vercel serverless app with lazy ML imports, in-memory SQLite fallback, and rate-limited REST API (SlowAPI).

Real-Time Credit Card Fraud Detection *Python · Scikit-learn · SHAP · Flask · Streamlit*

- ▶ Built a Random Forest classifier optimized for high recall to detect fraudulent transactions in real time, with SHAP-based explainability to surface the top contributing features per prediction for audit transparency.
- ▶ Deployed dual interfaces for production use — a REST `/predict` API for system integration and a Streamlit dashboard with live demo, audit logging, and metrics visualization.

EXPERIENCE

Cybersecurity Intern — Edureka (Remote)

Mar 2025 – Jul 2025

- ▶ Led a peer team through ethical hacking labs covering vulnerability scanning, password cracking, and red/blue team simulations as part of a structured offensive security curriculum.
- ▶ Completed hands-on training in network penetration testing and defensive countermeasures, applying techniques across multiple lab-based attack-and-defend simulations.

EDUCATION

B.Tech in Cybersecurity — NIIT University, Neemrana, Rajasthan

2022 – 2026

Coursework: Network Security · Ethical Hacking · Digital Forensics · Cryptography · OS · DBMS

Certifications: Google Cloud Career Practitioner Pathway (GDSC, 2025) · Cybersecurity & Ethical Hacking — Edureka (2025) · Intro to Cybersecurity — Cisco (2024)